MIL–P–45605A(MU)
24 MARCH 1964
SUPERSEDING
MIL–B–45605(ORD)
24 FEBRUARY 1959

MILITARY SPECIFICATION

PLOTTING BOARDS, FLASH RANGING, FIRE CONTROL: M5A2E2 AND M18

1. SCOPE

1.1 This specification covers Flash Ranging Plotting Boards M5A2E2 and M18 which are used to plot the location of enemy batteries (see 6.1).

2. APPLICABLE DOCUMENTS

2.1 The following documents, of the issue in effect on date of invitation for bids or request for proposal, form a part of this specification to the extent specified herein.

SPECIFICATIONS

MILITARY

MIL–F–13926 —Fire Control Materiel; General Specification Governing the Manufacture and Inspection of.

MIL–P–14232 —Parts, Equipment and Tools for Ordnance Materiel, Packaging of.

MIL–I–45208 —Inspection System Requirements.

STANDARDS

MILITARY

MIL–STD–105 —Sampling Procedures and Tables for Inspection by Attributes.

MIL–STD–109 —Quality Assurance Terms and Definitions.

DRAWINGS

U.S. ARMY MUNITIONS COMMAND

F8567955—Plotting Board, Flash Ranging, Fire Control: M18.

F8582661—Plotting Board, Flash Ranging, Fire Control: M5A2E2.

(Copies of specifications, standards, drawings, and publications required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

3. REQUIREMENTS

3.1 Materials. Materials shall be in accordance with drawings, material specifications, and general specifications forming a part of this specification.

3.2 Fabrication. The plotting boards shall be manufactured in accordance with Drawing F8582661 for the board M5A2E2 and F8567955 for the board M18 and the respective drawings pertaining thereto. The requirements of this specification are detailed only to the extent considered necessary to insure inherent quality and performance capabilities.

3.3 General specification. The contractor shall be responsible for adherence to, and compliance with, the following requirements of Specification MIL–F–13926.

(a) Order of precedence

(b) Dimensions and tolerances

FSC 1220

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(c) Inorganic protective surface finishes

(d) Part identification and marking

(e) Workmanship

3.4 **Environmental.**

3.4.1 *Storage temperatures.* The plotting boards shall show no evidence of physical failure and shall meet all performance requirements of this specification at a standard ambient temperature of between 60° and 90° F. after having been exposed to and thermally stabilized at ambient temperatures of −80° and +160° F.

3.4.2 *Operating temperatures.* The plotting boards shall meet the applicable requirements of 3.8 while exposed to and thermally stabilized at ambient temperatures of −40° and +150° F.

3.5 **Plotting scale flatness.** The plotting scale of the drafting machine shall lie flat from the "7" graduation to the "15" graduation when the "11" graduation is at the center of the board. The ends of the scale shall lie flat within 0.031 inch.

3.6 **Parallelism.** When alined with the fiducial edge of the plotting scale, the grid lines of the plotting disk shall be parallel to the fiducial edge within 0.015 inch throughout their entire length.

3.7 **Accuracy.** Lines drawn forming a centrally located square measuring 16,000 meters per side shall portray a perfect square. Each side shall be equivalent to 16,000 meters ±15 meters. Lines connecting the diagonally opposite corners of the square shall form the hypotenuse of right triangles. The hypotenuse of each shall measure 22,627 meters ±21 meters.

3.8 **Operability.**

3.8.1 *Torque.* The torque required to rotate the slow motion knob shall not exceed 9 pound inches at a standard ambient temperature of between 60° and 90° F., and 12 pound inches at the extreme operating temperatures specified in 3.4.2.

3.8.2 *Plotting board table.* With the brake lever pressed and held, it shall be possible to rotate the plotting board table through its entire travel without binding. With the brake lever released, the braking mechanism shall hold the table in a clamped position but will allow fine adjustment with the slow motion knob.

3.8.3 *Drafting machine.* There shall be no binding in any part of the drafting machine when the plotting scale of the machine is moved back and forth over the entire plotting disk.

3.8.4 *Plotting disk surface.* The plotting disk shall have a surface that will allow clear pencil markings and clean erasures. The surface shall have no defects such as pits, digs, scratches, and stains which may be mistaken for pencil marks.

4. QUALITY ASSURANCE PROVISIONS

4.1 **Responsibility for inspection.** Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified, the supplier may utilize his own or any other inspection facilities and services acceptable to the Government. Inspection records of the examinations and tests shall be kept complete and available to the Government as specified in the contract or order. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.1.1 *General provisions.* The quality assurance provisions of this specification and of other documents referenced herein form the basis for inspection to be performed by the supplier. Inspection shall be in accordance with the requirements of Specification MIL–I–45208 and the component inspection requirements of Specification MIL–F–13926. Definitions of inspection terms not otherwise defined herein shall be as listed in Standard MIL–STD–109.

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4.2 Inspection provisions.

4.2.1 *Inspection lot.* Unless otherwise specified by the contracting officer, inspection lot size, formation, and presentation of lots shall be in accordance with "Submission of Product" and "Drawing of Samples" specified in Standard MIL–STD–105.

4.2.2 *Examination and tests.* Components and subassemblies shall be inspected in accordance with the inspection provisions contained in Specification MIL–F–13926. Examinations and tests related to section 3 herein shall be performed on a defect (individual characteristic) basis in accordance with Standard MIL–STD–105, and the inspection level and sampling plans specified in table I titled "Classification of Defects." Examinations and tests for Packaging, Packing and Marking shall be in accordance with Specification MIL–P–14232 and section 5 herein. The tabulated classification of defects shall constitute the minimum inspection to be performed by the supplier prior to Government acceptance or rejection by lot. The Government reserves the right to inspect for any applicable requirement, and to reject individual nonconforming items.

TABLE I. *Classification of defects*
Use Level II of table I with Sampling Plan table II–A of Standard MIL–STD–105.

	Requirement	Test procedure
Critical: None		
Major: AQL 1.0%		
101. Plotting scale flatness	3.5	4.5.3
102. Parallelism	3.6	4.5.4
103. Accuracy	3.7	4.5.5
104. Torque	3.8.1	4.5.6
105. Plotting board table	3.8.2	Tactile
106. Drafting machine	3.8.3	Tactile
Minor: AQL 2.5%		
201. Plotting disk surface	3.8.4	Visual

4.2.3 *Disposition of non-conforming product.* Rejected lots shall be screened for all defective characteristics. Removal or correction of defective units and resubmittance of rejected lots shall be in accordance with "Acceptance and Rejection" as specified in Standard MIL–STD–105.

4.3 Control tests.

4.3.1 *General.* One plotting board shall be selected at random from each 100 produced, or from each month's production, whichever occurs first. Control samples shall have successfully met all acceptance tests specified herein prior to conducting the following examinations and tests:

Control test	*Requirement*	*Test procedure*
Materials	3.1	Drawings and Specifications
Fabrication	3.2	Visual and Drawings
General Specification	3.3	Visual and MIL–F–13926

4.3.2 *Temperature tests.* Three plotting boards shall be selected at random from each 50 produced, or from each month's production, whichever occurs first, to determine compliance with the storage and operating temperatures as specified in 3.4.1, 3.4.2, 4.5.1 and 4.5.2.

4.3.3 *Control test failure.* Should a control test sample fail to meet the specified test requirements, acceptance of the product will be suspended by the Government until necessary corrections have been made by the contractor and the resubmitted samples have been approved.

4.4 Inspection equipment. Except as otherwise provided for by the contract, the contractor shall furnish and maintain all required measuring and testing equipment in accordance with the "Measuring and Test Equipment" requirement of Specification MIL–I–45208. The Government reserves the right to use the test equipment for its own independent inspections to the extent that such use will not unduly interfere with the contractor's delivery schedule.

4.4.1 *Government furnished inspection equipment.* Where the contract provides for Government furnished test equipment, care and maintenance of test equipment shall be in accordance with "Measuring and Test Equipment" and "Government Furnished Material" requirements of Specification MIL–I–45208.

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4.4.2 *Contractor furnished inspection equipment.*

4.4.2.1 *Government design.* All inspection equipment required by drawings forming a part of the contract and not provided by the Government shall be supplied by the contractor in accordance with the "Inspection Provisions" of Specification MIL–I–45208.

4.4.2.2 *Contractor design.* The contractor shall design and supply inspection equipment compatible with the "Test Methods and Procedures" specified in 4.5, and with the "Component Inspection" requirements of Specification MIL–F–13926. Concept, construction, materials, dimensions, and tolerances used in design of test equipment shall be so selected and controlled as to insure that the test equipment will permit positive rejection of a product which exceeds the prescribed tolerance limit, and will reliably indicate acceptability of a product which does not exceed 90% of the prescribed tolerance limit. Construction shall be such as to facilitate routine calibration of the equipment.

4.5 Test methods and procedures.

4.5.1 *Storage temperature.* The storage temperature for the plotting boards at minus 80 degrees and plus 160 degrees Fahrenheit shall be conducted in conformance with 3.4.1 and the "Test Facilities" of Specification MIL–F–13926. Upon return to standard ambient temperature the plotting boards shall be subjected to a visual and tactile examination and shall show no evidence of physical failure such as cracks, splits, or warpage. The plotting boards shall then be subjected to all the tests specified herein.

4.5.2 *Operating temperature.* The temperature cycling for the operability tests of the plotting boards at the specified temperature extremes shall be conducted in accordance with 3.4.2 and the "Test Facilities" of Specification MIL–F–13926. When the plotting boards are thermally stabilized at minus 40 degrees and plus 150 degrees Fahrenheit, respectively, they shall be subjected to the operability tests specified in 4.5.6.

4.5.3 *Plotting scale flatness.* This test shall be performed by placing the "11" graduation of the plotting scale at the center of the board, and measuring any variation in flatness at the "7" and "15" graduations of the scale with standard thickness gages. Any error in flatness shall not exceed the requirements of 3.5.

4.5.4 *Parallelism.* Parallelism of the grid lines in relation to the fiducial edge of the plotting scale shall be tested on the two grid lines which intersect through the center of the plotting disc. (Rotate the table 1600 mils to obtain alinement of the perpendicular grid line with the plotting scale.) Use a pocket optical comparator to check errors in parallelism. Errors shall not exceed the requirements of 3.6.

4.5.5 *Accuracy.* The fiducial edge of the scale of the drafting machine shall be set to coincide with the north-south grid lines so that it can be moved back and forth across the plotting surface remaining parallel to its original position. With the fiducial edge of the scale in this position, a point shall be established approximately in the center of the plotting board. Using the grid squares as units of measure, a point shall be marked off 8,000 meters each side of the established center, on the east-west grid line and a vertical line shall be drawn through each of these points. Rotate the grid plotting board 1600 mils in a clockwise direction following the same procedure on the new east-west grid line. Extend all lines to complete a square. Rotate plotting board counterclockwise 800 mils and draw a vertical line bisecting the square into two triangles. Rotate the plotting board counterclockwise 1600 mils and repeat performance. These new lines shall form a hypotenuse in compliance with 3.7 as measured with the plotting scale.

4.5.6 *Torque.* This test shall be performed using a commercial type inch-pound torque wrench with adapter to accommodate the slow motion knob. Readings with the torque wrench shall be taken in both directions while in motion. The torque required to rotate the slow motion knob at standard ambient temperature and at the extreme temperatures of

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3.4.2 shall not exceed the values as specified in 3.8.1.

5. PREPARATION FOR DELIVERY

5.1 Packaging, packing, and marking. Packaging, packing, and marking shall be in accordance with Specification MIL-P-14232 and the applicable Packaging Data Sheet. The level of protection shall be as specified in the procurement document.

6. NOTES

6.1 Intended use. Flash Ranging Plotting Boards M5A2E2 and M18 are used to determine the location of hostile batteries by plotting the line of azimuth of the flash or smoke from the weapons as reported by two or more observation posts. Plotting Board M18 may also be used for sound ranging through installation of a supplementary scale and index tabs.

6.2 Ordering data. Procurement documents should specify the following:

(a) Title, number, and date of this specification.
(b) Selection of applicable levels of preservation, packaging, and packing.
(c) Applicable stock number.
(d) Applicable Packaging Data Sheet (see 5.1).

Custodian:
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Project No. 1220-A150Y

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SPECIFICATION ANALYSIS SHEET

Form Approved
Budget Bureau No. 119-R004

INSTRUCTIONS

This sheet is to be filled out by personnel either Government or contractor, involved in the use of the specification in procurement of products for ultimate use by the Department of Defense. This sheet is provided for obtaining information on the use of this specification which will insure that suitable products can be procured with a minimum amount of delay and at the least cost. Cooperation and the return of this form will be appreciated. Fold edges on reverse side, staple in corner, and send to preparing activity (see indicated on reverse hereof).

SPECIFICATION

ORGANIZATION (Of submitter) | CITY AND STATE

CONTRACT NO. | QUANTITY OF ITEMS PROCURED | DOLLAR AMOUNT
$

MATERIAL PROCURED UNDER A

☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

1. HAS ANY PART OF THE SPECIFICATION CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?

A. GIVE PARAGRAPH NUMBER AND WORDING.

B. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES.

2. COMMENTS ON ANY SPECIFICATION REQUIREMENT CONSIDERED TOO RIGID

3. IS THE SPECIFICATION RESTRICTIVE?

☐ YES ☐ NO IF "YES", IN WHAT WAY?

4. REMARKS (Attach any pertinent data which may be of use in improving this specification. If there are additional papers, attach to form and place both in an envelope addressed to preparing activity)

SUBMITTED BY (Printed or typed name and activity) | DATE

DD FORM 1 APR 63 1426 REPLACES NAVSHIPS FORM 4869, WHICH IS OBSOLETE PLATE NO. 15419

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